

Assignment-cum-project

15 Marks

Students need to see the dataset carefully located on the Google Drive (https://drive.google.com/drive/folders/1mvmW-KQ7f0w2MnO7NkNUWF6bF7LG3v1d?usp=drive_link). It is a time series noise-removed (sampling rate:100 Hz) dataset. Data is highly skewed, i.e., imbalanced (binary class), which generally impacts false positives and false negatives while classifying. Students need to handle the imbalance in the data before classifying. But some restrictions need to be maintained: **any of the minority class (i.e., '1') data should not be removed, nor should it be up-sampled.** The group may use any data imbalance handling algorithm to make the final result better. Students should report the classification results only on the following metrics: True positive (TP), True Negative (TN), False Positive (FP), False Negative (FN), F1 score, and balanced accuracy. Each group can use any 4 (four) supervised classifiers (shallow/deep/ensemble) for comparative analysis. Students should report the results in both cases: without using any data balancing technique and with using a data balancing technique. They should use the Stratified k-fold cross-validation technique with a set of hyperparameter tuning relevant to the corresponding classifiers. For scanning the data, a **sliding window** (length 50 samples) technique should be used (50% overlap ratio). A window will be labelled as '1' if more than 40% of samples are '1', otherwise it will be labelled as '0'. The final classification will be based on the window labels.

Data Description: Each folder contains a set of .csv files. Each file has 9 features (the first two columns are not features) and one label (the last column). Each row is labelled as either '0' or '1'. Each window will be treated as a sample. A file may have both classes or only the majority class (i.e., '0'). Just one sample for the training and testing division has been given. Each group will take a different sample set (exactly 6 folders) for testing, **but remember that the test set should have both classes.**

Performance will be assessed based on the code and quality of the result (raw confusion matrix). Preference will be given for writing the code from scratch, specifically for the ML model and its performance evaluation. Code should have multiple modules/functions as per the need.

Note: Students can discuss this problem-related issues (if any) during the classes or coming to my cabin (FS 12, 1st floor, R & D building). Each group will submit the code and corresponding o/p (might be graphical) to the Google Drive in a **separate folder for each group (shareable to that particular group's students and the teacher only)**.

Deadline: 31st March 2026